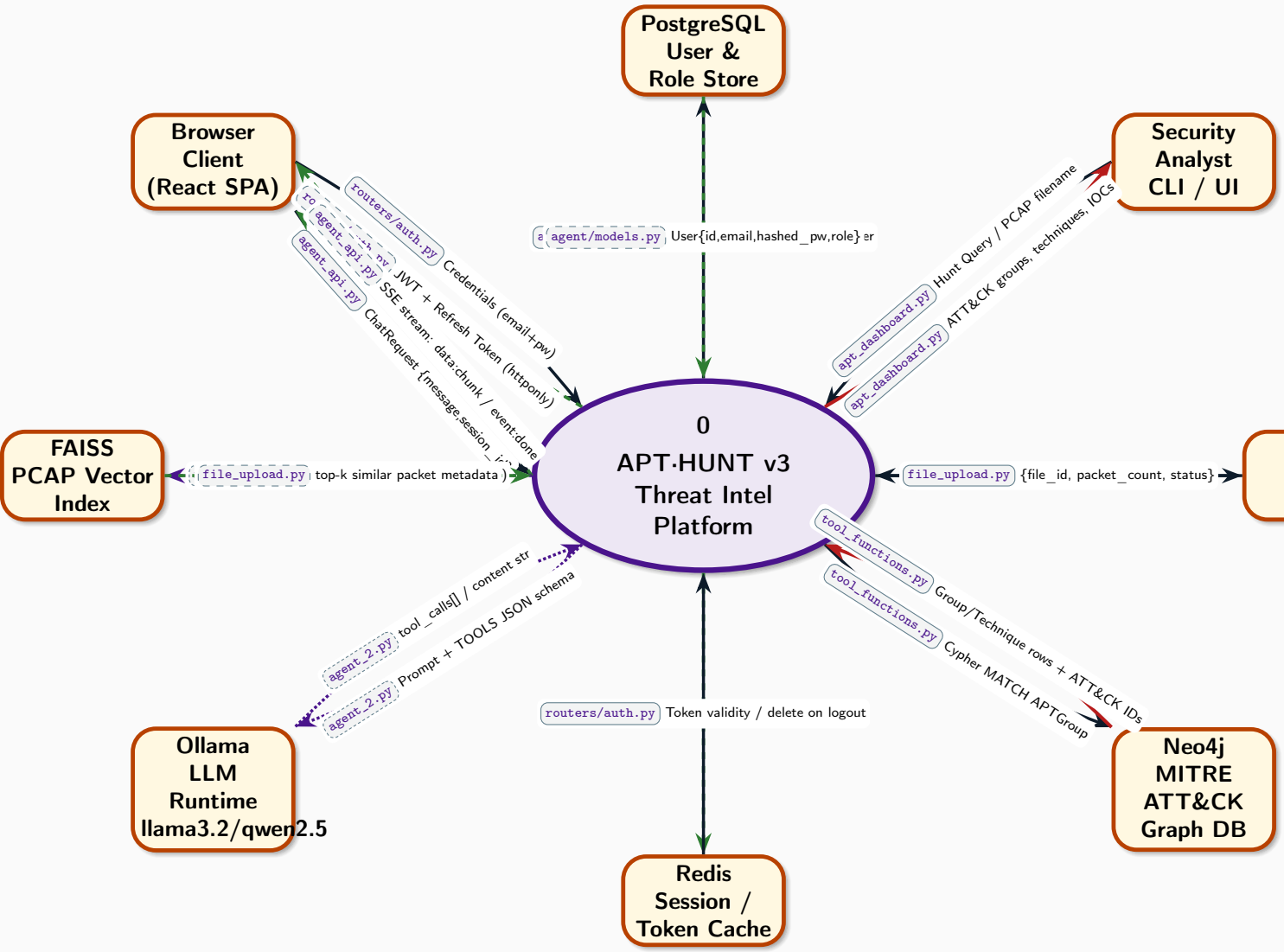
DFD Level 0 Context Diagram

Every external entity and labelled flow maps directly to a source module in the repository.

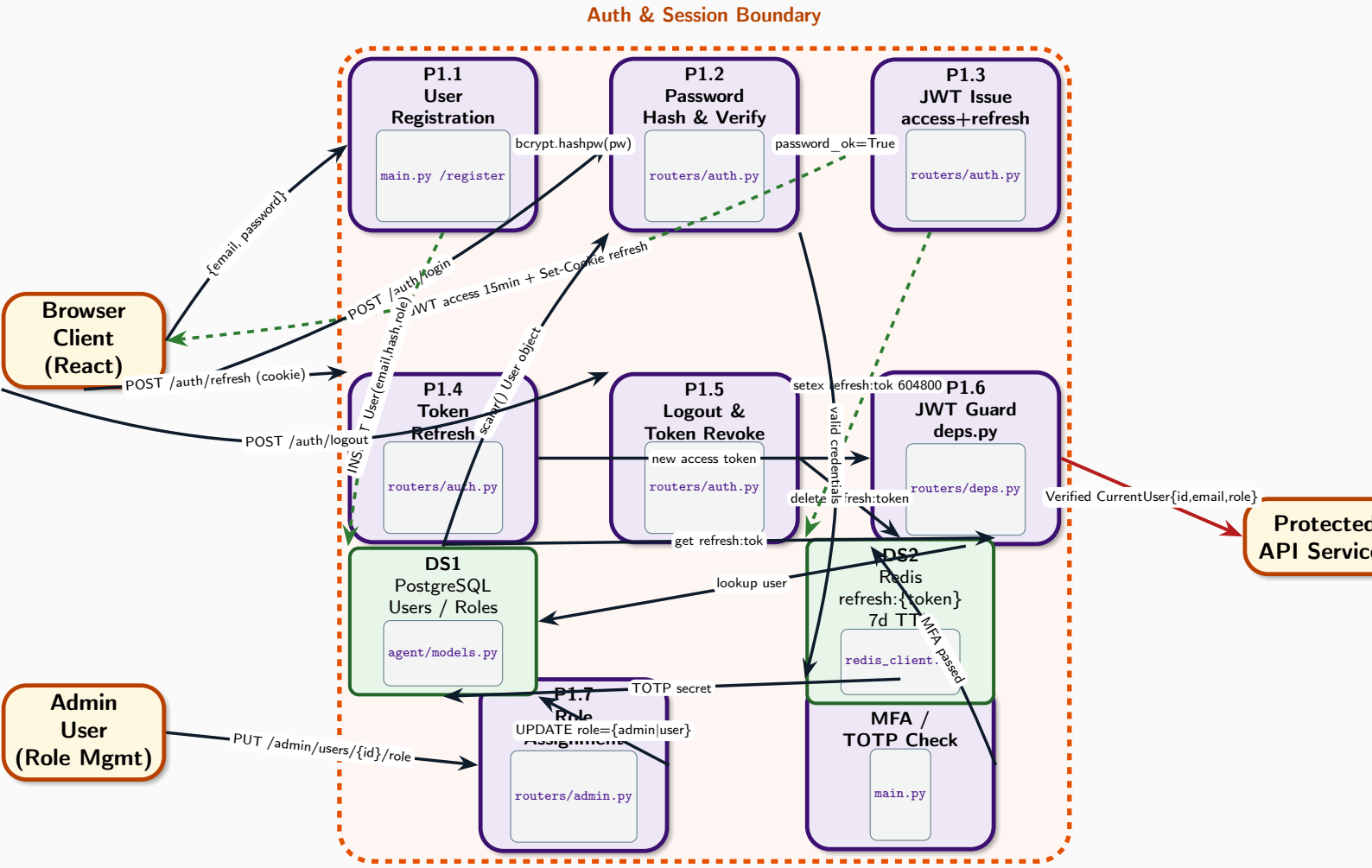


Flow Legend

- REST / internal call
- Encrypted (JWT/bcrypt/TLS)
- Alert / Threat / ATT&CK
- AI / LLM inference

DFD Level 1(A) Authentication & Session Management

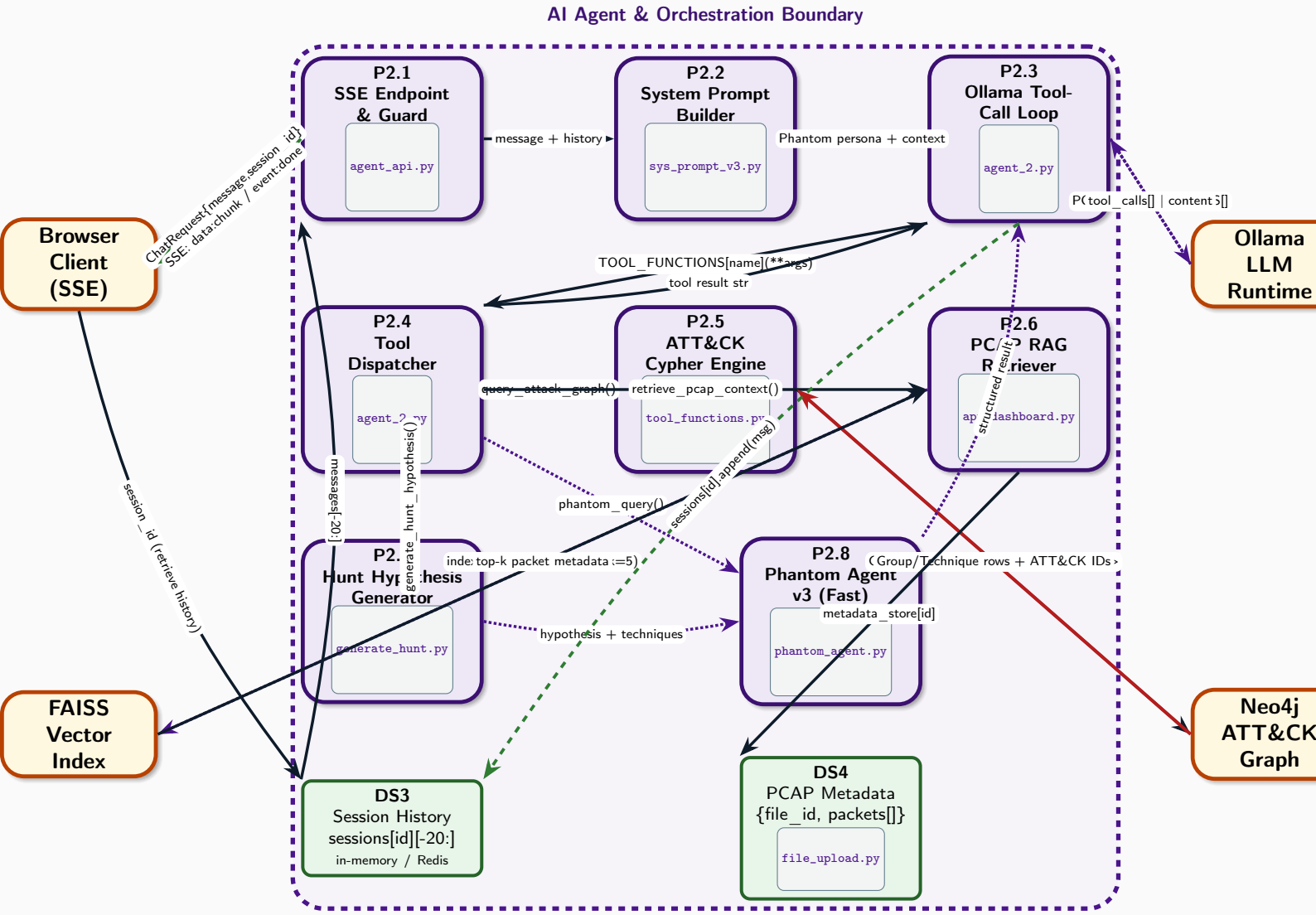
Modules: main.py, routers/auth.py, routers/deps.py, routers/admin.py, agent/db.py, redis_client.py



DFD Level 1(B) AI Agent Orchestration & Tool Loop

APT-HUNT v3

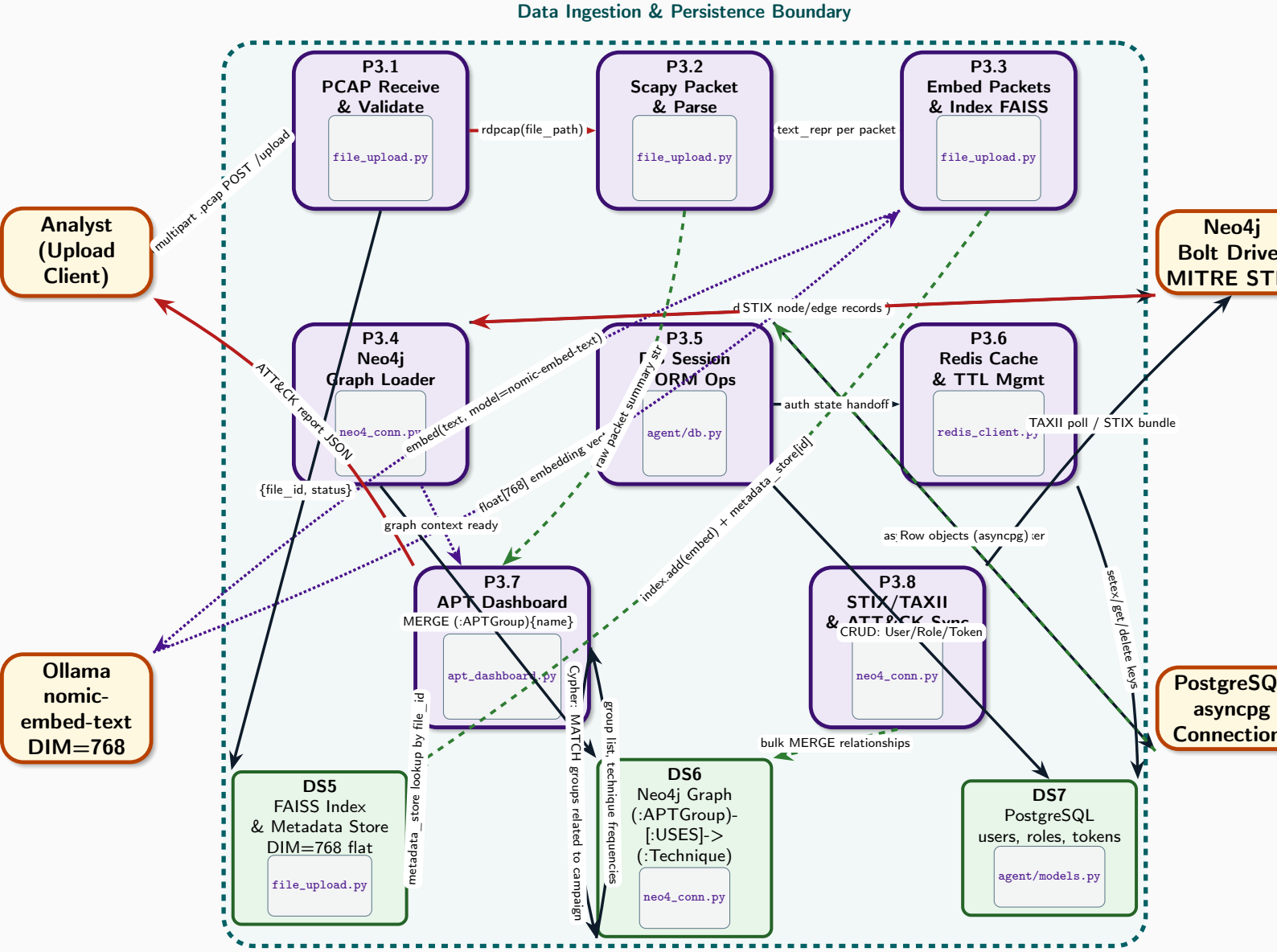
Modules: agent_api.py, agent_2.py, phantom_agent.py, tool_functions.py, sys_prompt_v3.py, generate_hunt_hypothesis.py



DFD Level 1(C) Data Ingestion, Storage & Persistence

Modules: `file_upload.py`, `neo4_conn.py`, `agent/db.py`, `redis_client.py`, `apt_dashboard.py`

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Data Store & Process Master Index — All Three L1 Diagrams

DFD-A	Auth & Session	DFD-B	AI Agent Loop	DFD-C	Ingestion & Storage
DS1	PostgreSQL Users/Roles	DS3	Session history (Redis/mem)	DS5	FAISS index + metadata
DS2	Redis refresh tokens (7d)	DS4	PCAP metadata store	DS6	Neo4j ATT&CK graph
				DS7	PostgreSQL ORM store
`routers/auth.py`	Login/Refresh/Logout	`agent_2.py`	Ollama tool-call loop	`file_upload.py`	Scapy + FAISS ingest
`routers/deps.py`	JWT guard / CurrentUser	`tool_functions.py`	Cypher dispatchers	`neo4_conn.py`	STIX/TAXII graph sync
`routers/admin.py`	Role assignment	`phantom_agent.py`	Fast concurrent agent	`apt_dashboard.py`	ATT&CK report gen